
Case Study 3:

Modeling and Control of Heart Rate Response to Treadmill Exercise

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References

Material presented in this case study was taken from:

- S. W. Su, L. Wang, B. G. Celler, A. V. Savkin, and Y. C. Guo, “Identification and control for heart rate regulation during treadmill exercise,” *IEEE Transactions on Biomedical Engineering* Vol. 54, No. 7, pp. 1238-1246, 2007.
- T. M. Cheng, A. V. Savkin, B. G. Celler, S. W. Su, and L. Wang, “Nonlinear modelling and control of human heart rate response during exercise with various work load intensities,” *IEEE Transactions on Biomedical Engineering*, Vol. 55, No. 11, pp. 2499-2508, 2008.

Objectives

- To propose a simple dynamic model that describes the heart rate response to treadmill walking exercise.
- To develop a computer-controlled treadmill system for the regulation of heart rate during treadmill exercise.

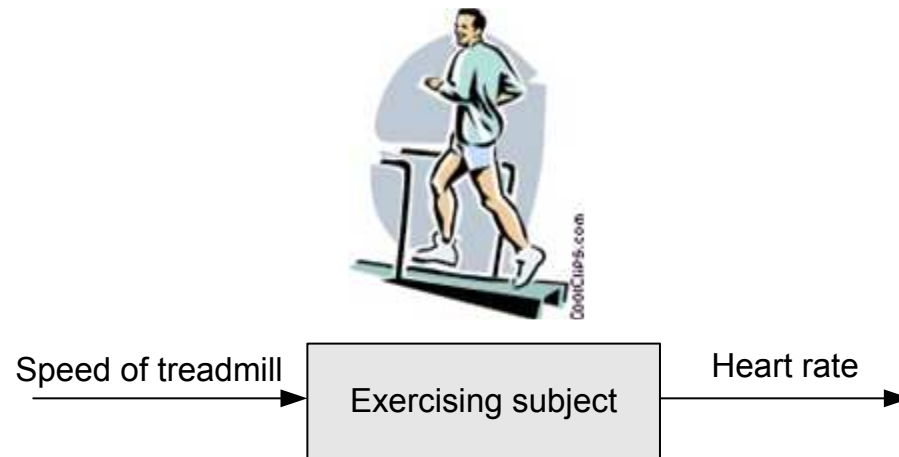


Motivation

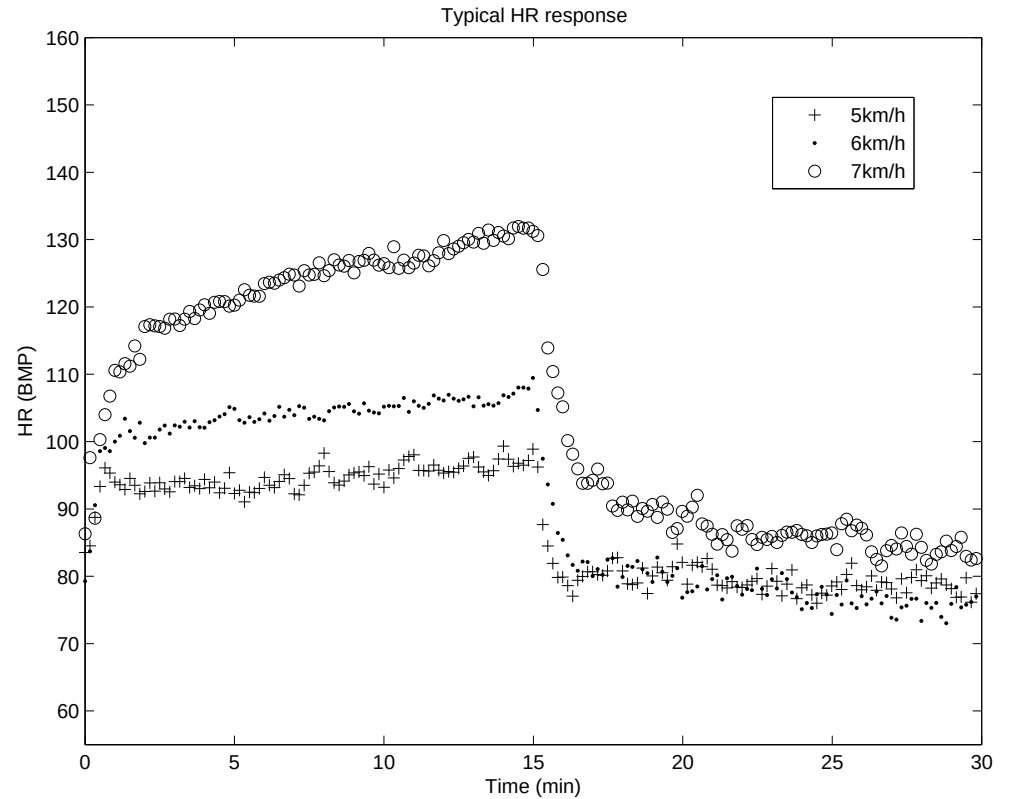
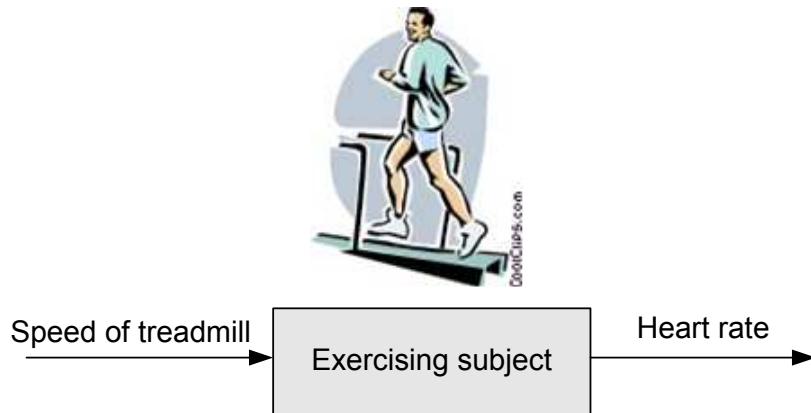
- Monitoring heart rate provides a convenient way of estimating oxygen consumption and energy expenditure during exercise.
- Understanding how heart rate responding to exercise will lead to an improvement:
 - in developing training protocols for athletics and more efficient weight loss protocols for the obese;
 - in facilitating assessment of physical fitness and health of individuals.
- Understanding the heart rate response during, and recovery after, an exercise, may also be beneficial to predicting cardiovascular disease mortality.
- Knowing the cardiovascular system responding to the stress induced by physical exercise provides us another perspective on how this system functions. For instance, this may give us some measures for the prevention of cardiac failure from hemodialysis.

Modeling

● Part 1: Heart Rate Response Modeling



Typical HR responses



- Responses are *not* linear!
- Require nonlinear model.

The Proposed Model

- We propose the following simple nonlinear control systems to model the heart rate (HR) response to treadmill walking exercise:

$$\begin{aligned}\dot{x}_1(t) &= -a_1x_1(t) + a_2x_2(t) + a_2s^2(t) \\ \dot{x}_2(t) &= -a_3x_2(t) + \phi(x_1(t)) \\ z(t) &= x_1(t)\end{aligned}\tag{1}$$

where

$$\phi(x_1(t)) := \frac{a_4x_1(t)}{1 + \exp(-(x_1(t) - a_5))}$$

- Output: $z(t)$, representing the change in heart rate (bpm) from rest.
- Model parameters: $a_1, \dots, a_5$ are positive scalars.
- The control input $s(t)$ (km/h) represents the speed of the treadmill.

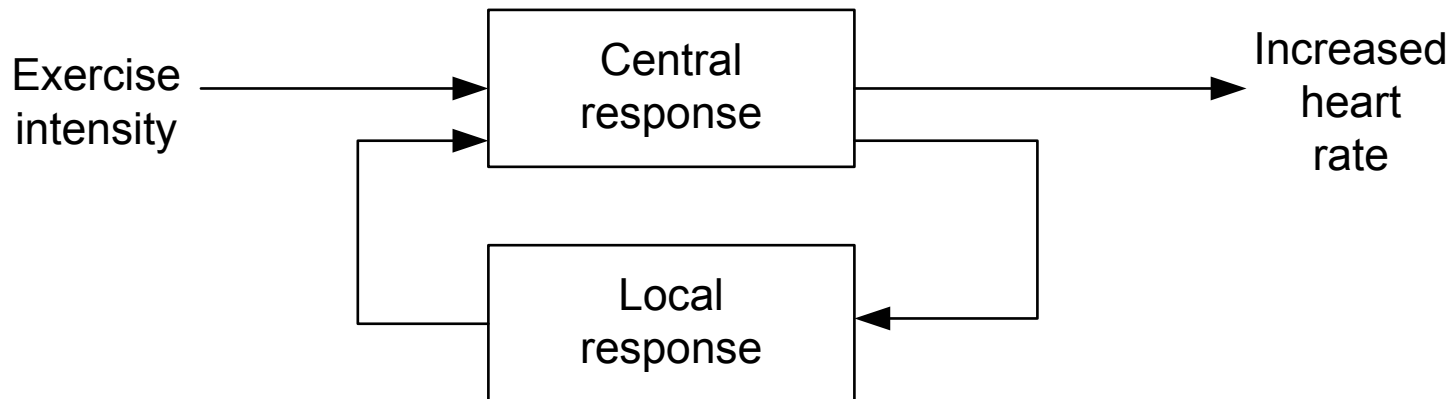
The Model (Cont'd)

$$\dot{x}_1(t) = -a_1x_1(t) + a_2x_2(t) + a_2s^2(t)$$

$$\dot{x}_2(t) = -a_3x_2(t) + \phi(x_1(t))$$

$$z(t) = x_1(t)$$

- $x_1(t)$ describes the change of HR due to the **central response**.
- $x_2(t)$ describes the complex **slow-acting local effects** from, e.g. the hormonal systems, the peripheral local metabolism, and /or the increase in body temperature, etc..



Experimental Verification

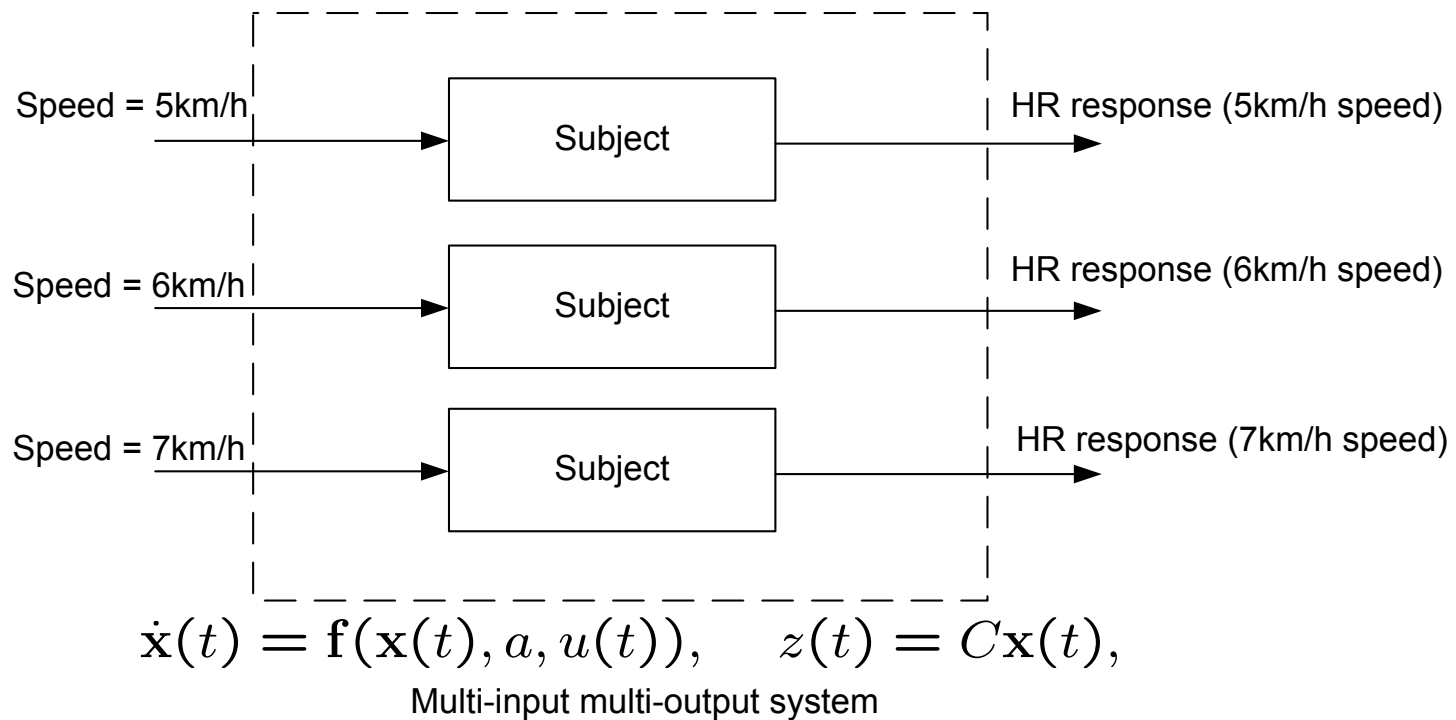
- **Subjects:** Six healthy male subjects were studied.
- **Procedure:** Each subject completed 3 exercise sessions. In each session, a subject was requested to walk on a treadmill at a given speed (5km/h, 6km/h, and 7km/h) for 15 minutes with a recovery period of 20 minutes.



Figure 1: Treadmill walking exercise.

Parameter Estimation

- The parameters of the model were estimated by using the 3 sets of input-output measurements of the *average responses* from the experiments.



Parameter Estimation (Cont'd)

- Therefore, we considered it as a parameter estimation problem of the multi-input-multi-output system:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), a, u(t)), \quad z(t) = C\mathbf{x}(t), \quad \mathbf{x}(0) = 0. \quad (2)$$

- $a = [a_1 \ a_2 \ \dots \ a_5]^T \in \mathbb{R}^5$ is the parameter vector.
- $u = [u_1 \ u_2 \ u_3]^T \in \mathbb{R}^3$ is the vector of 3 treadmill speeds
- $z = [z_1 \ z_2 \ z_3]^T$.
- $\mathbf{x} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3]^T \in \mathbb{R}^6$.
- For $i = 1, 2, 3$, the vector $\mathbf{x}_i := [x_{i,1} \ x_{i,2}]^T$ and y_i are the state vector and the output from the input u_i , respectively.
- The Least squares method was then applied to determine an estimate of a .

Parameter Estimation (Cont'd)

- To make the estimation process more robust:
 - the unit of time t in system (2) was defined in minute.
 - scalings were also carried out for the control input and the output.
 - The speed of the treadmill were normalized by 8 km/h (assumed the max. walking speed was 8 km/h).

$$u(t) := [s_1(t) \ s_2(t) \ s_3(t)]' / 8. \quad (3)$$

Three constant speeds were used: $s_1 = 5\text{km/h}$, $s_2 = 6\text{km/h}$ and $s_3 = 7\text{km/h}$.

- The output $z_i(t)$ from the input $u_i(t)$ was defined as

$$z_i(t) := (\text{HR}_i(t) - 73.4) / 4, \quad (4)$$

where $\text{HR}_i(t)$ is the absolute heart rate at time t , 73.4bpm is the mean resting heart rate, and 4 is the scaling factor.

Parameter Estimation (Cont'd)

- The least squares method was then employed for the parameter estimation. The objective function was in the form

$$S(a) = \sum_{i=1}^N (z(t_i) - \hat{z}(t_i, a))' Q (z(t_i) - \hat{z}(t_i, a)) \quad (5)$$

where, for $i = 1, 2, \dots, N$, $z(t_i)$ is the measurement of the output vector at time t_i , $\hat{z}(t_i, a)$ is the output of system (2) with the parameter vector a , and Q is the weighting matrix. In this study,

$$Q = \begin{bmatrix} 2.5 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ was used. With the objective function (5), the}$$

least squares method was used to determine an estimate of a which was denoted as $\hat{a} := [\hat{a}_1 \ \hat{a}_2 \ \dots \ \hat{a}_5]'$.

Parameter estimates

Parameter estimates (Confidence intervals, δa)				
$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$	$\hat{a}_4$	$\hat{a}_5$
1.857 (0.119)	24.812 (1.479)	5.75×10^{-2} (6.89×10^{-3})	3.05×10^{-3} (2.23×10^{-4})	8.254 (0.137)

Table 1: Estimated parameter values for the averaged responses of six subjects.

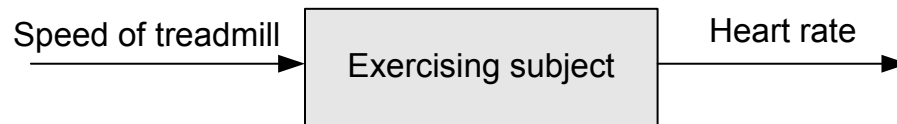
The estimated model

By combining (1), (3) and (4) with the estimated parameters, we obtained a *nonlinear* model to describe the heart rate $HR(t)$ (bpm) during and after treadmill exercise and it is given as follows:

$$\begin{aligned}\dot{x}_1(t) &= -1.857x_1(t) + 24.812x_2(t) + 0.388s^2(t) \\ \dot{x}_2(t) &= -5.75 \times 10^{-2}x_2(t) + \phi(x_1(t)) \\ HR(t) &= 4.0x_1(t) + HR_{\text{rest}}\end{aligned}\tag{6}$$

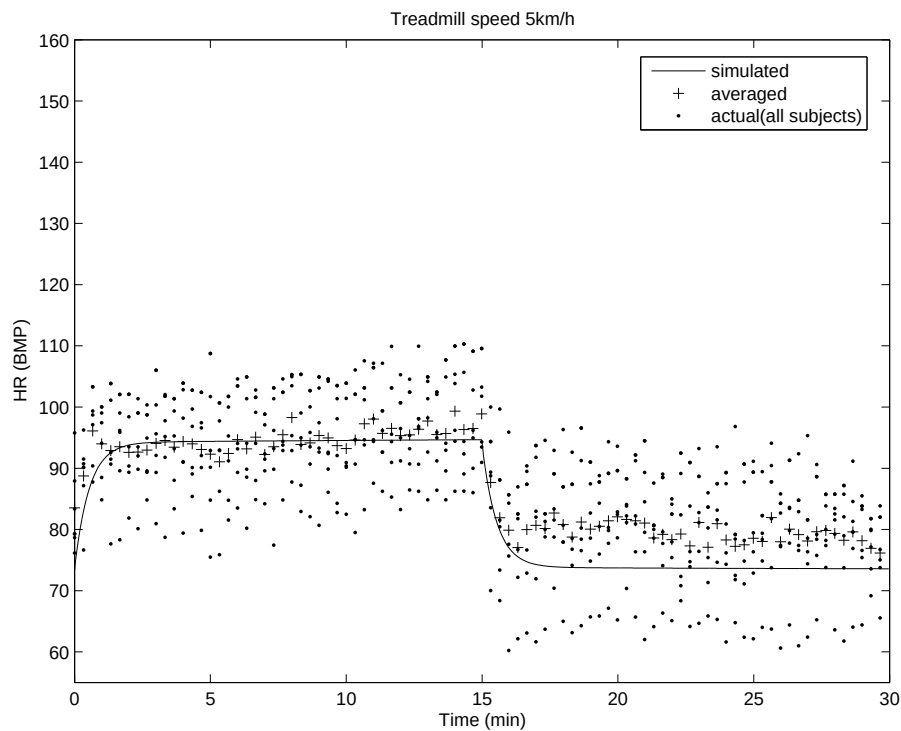
where $x_1(0) = x_2(0) = 0$, $s(t)$ is the speed of treadmill in km/h, the unit of time t is in minute, and

$$HR_{\text{rest}} := 73.4, \quad \phi(x_1(t)) := \frac{3.05 \times 10^{-3}x_1(t)}{1 + \exp(-(x_1(t) - 8.254))}.$$

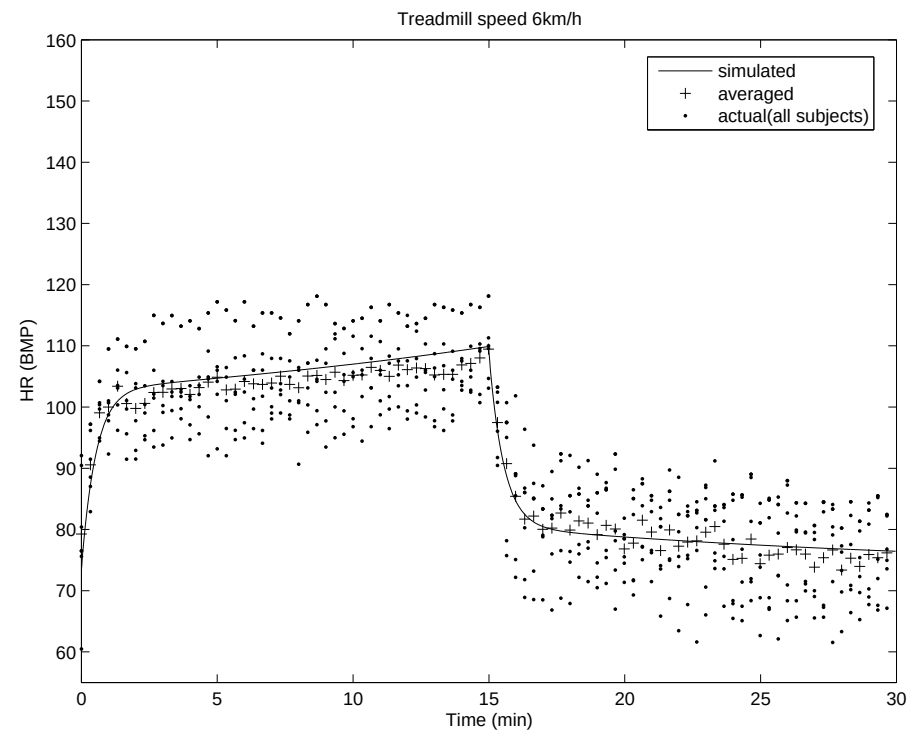


Modeling Results

Heart rate responses: actual responses from all 6 subjects (dots), average response (crosses) and simulated response (solid line)



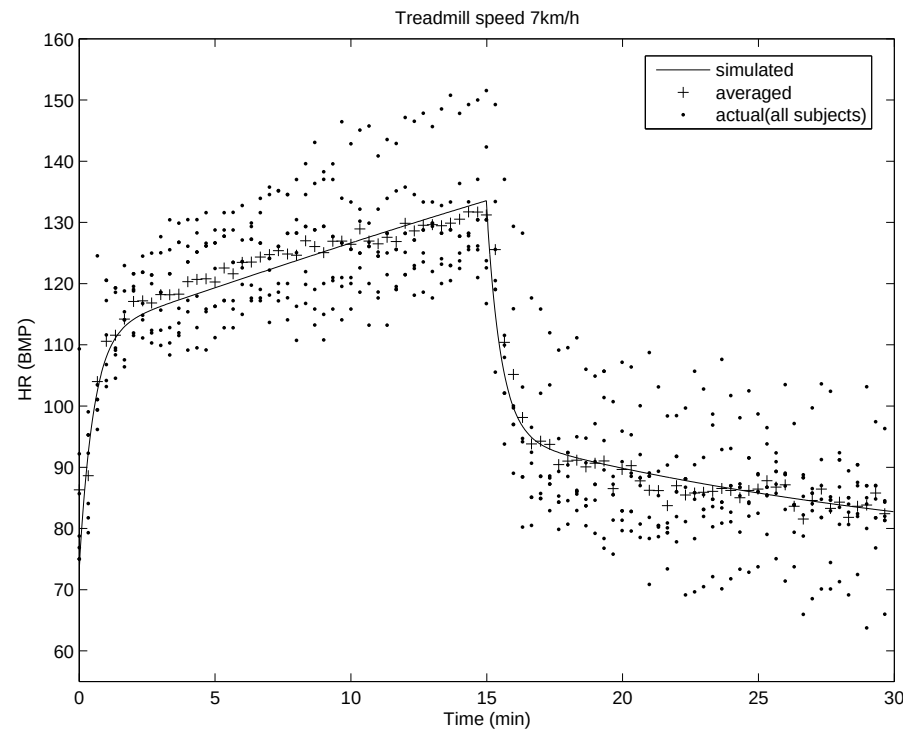
5km/h treadmill speed



6km/h treadmill speed

Modeling Results (Cont'd)

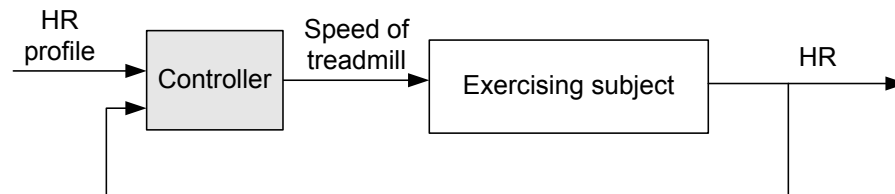
Heart rate responses: actual responses from all 6 subjects (dots), average response (crosses) and simulated response (solid line)



7km/h treadmill speed

Controller Design

● Part 2: Controller Design



A Nonlinear Control System

With proper scaling (since scaling makes model analysis and controller design easier), system (6) is written in a state-space form as follows:

$$\begin{aligned}\dot{\eta}(t) &= A\eta(t) + B_1\Phi(\eta_1(t)) + B_2g(u(t)) \\ y(t) &= C\eta(t)\end{aligned}\tag{7}$$

$$\begin{aligned}A &= \begin{bmatrix} -0.031 & 0.0276 \\ 0 & -0.001 \end{bmatrix}, \quad \eta = \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0.0276 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 \\ 0 \end{bmatrix}', \\ \Phi(\eta_1(t)) &:= \frac{7.5 \times 10^{-4}\eta_1(t)}{1 + \exp(-15(\eta_1(t) - 5.503 \times 10^{-1}))}, \quad g(u(t)) := u^2(t).\end{aligned}\tag{8}$$

To obtain (7), we have defined the normalized output

$y(t) = \eta_1(t) := x_1(t)/15$ (i.e. $y(t) := (\text{HR}(t) - \text{HR}_{\text{rest}})/60$) and the normalized input $u(t) := s(t)/8$. Also, for simplicity, the time unit of t in (7) is in second, instead of minute as in (6).

A Nonlinear Control System (Cont'd)

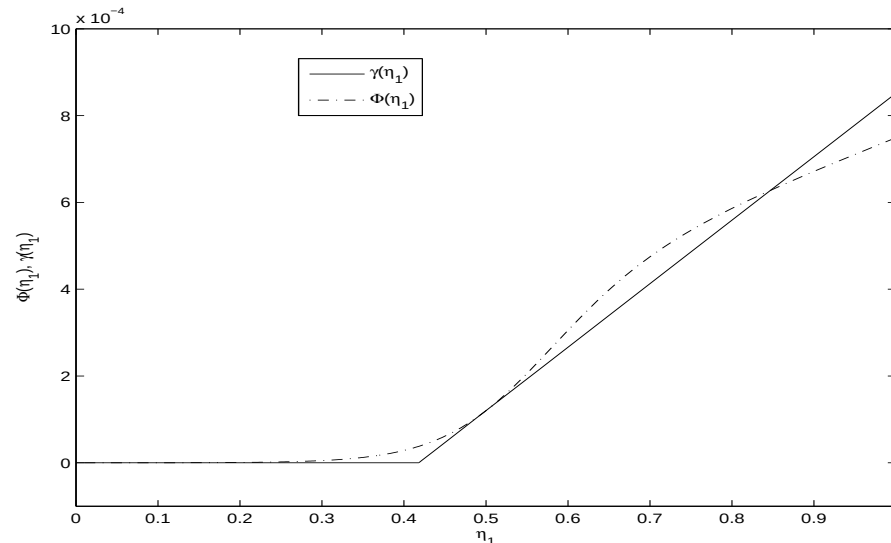
System (7) has nonlinearity $\Phi(\eta_1)$ and nonlinear control input $g(u)$.

- To overcome the control input nonlinearity, a transformed input $v(t) = g(u(t)) = u^2(t)$ is defined.

- $\Phi(\eta_1)$ can be approximated by a piecewise linear function

$$\gamma(\eta_1) = \begin{cases} 0 & \text{if } \eta_1 \leq 0.418 \\ 1.464 \times 10^{-3} \eta_1 - 6.119 \times 10^{-4} & \text{if } \eta_1 > 0.418. \end{cases}, \text{ is obtained}$$

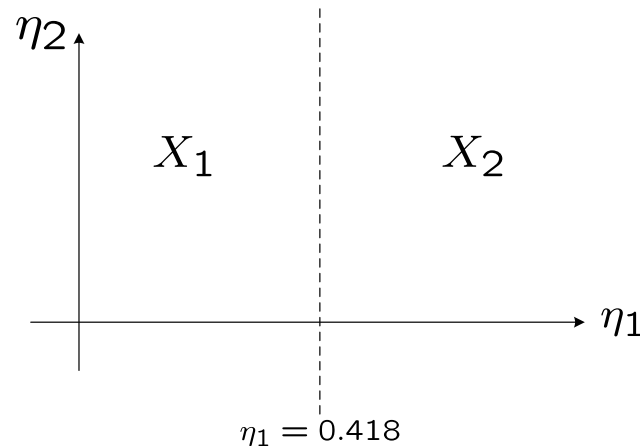
by linearizing at $\Phi(\eta_1)$ at $\eta_1 = 0$ and 0.5



A Nonlinear Control System (Cont'd)

- Define two partitions of the state space:

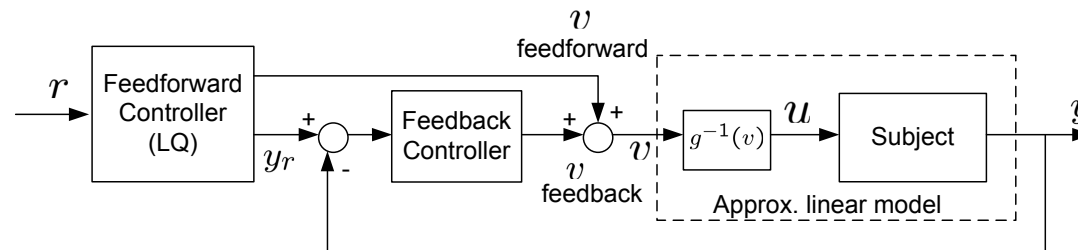
$$\begin{aligned} X_1 &:= \{[\eta_1 \ \eta_2]' \in \mathbb{R}^2 \mid \eta_1 < 0.418\}, \\ X_2 &:= \{[\eta_1 \ \eta_2]' \in \mathbb{R}^2 \mid \eta_1 \geq 0.418\}. \end{aligned} \tag{9}$$



- In each partition X_i , $i = 1, 2$, the nonlinear system $\dot{\eta} = f(\eta, v)$ can be approximated by a linear system.
- Therefore, we can design a linear controller in each partition $X_i, i = 1, 2$.

Controller Structure (2-DOF)

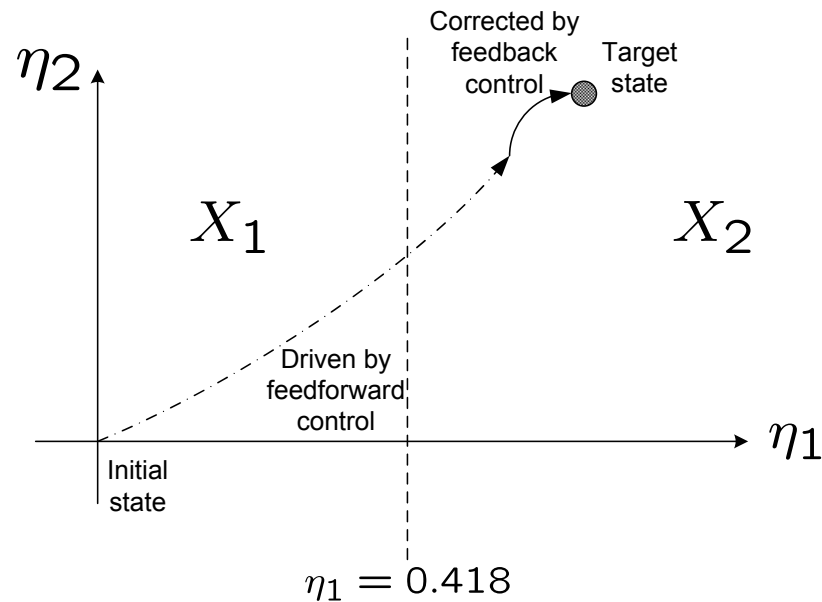
- Adopt a *two-degree-of-freedom (2-DOF)* controller, consisting of a *feedforward*(FF) controller and a *feedback*(FB) controller for the control and regulation of heart rate responses.



- The control $v = v_{feedforward} + v_{feedback}$.
- Feedback-reduces the effect of uncertainty (disturbances and model errors).
- Feedforward-shapes the command r to improve tracking performance.
- Design the FF and FB controllers separately.

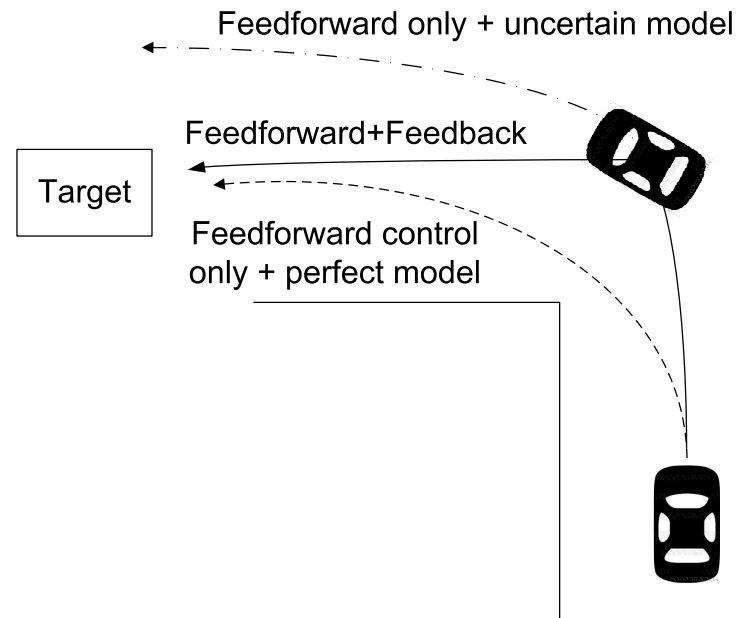
Controller Structure (2-DOF) (Cont'd)

- The feedforward controller (servo) drives the system from region X_1 to region X_2 .
- The feedback controller (regulator) *regulates* the system to the target state in region X_2 .



Controller Structure (2-DOF) (Cont'd)

- A simple example of 2DOF control: driving car round a corner.



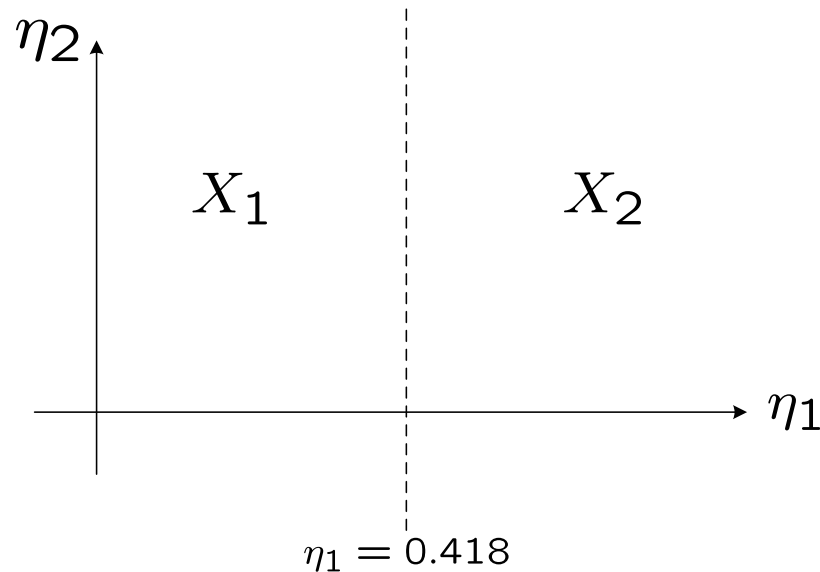
- A Feedforward control with a *perfect model* (and no disturbances) can reach target *without feedback*.
- A Feedforward control with an *uncertain model* cannot reach target.
- A feedforward control + feedback control with an *uncertain model* can reach target.

Feedforward (LQ) Controller

- Linearize the nonlinear model $\dot{\eta} = f(\eta, v)$ at $\eta_0 = [0.0 \ 0]'$ and $v_0 = 0.0$, we have the linearized model

$$\begin{aligned}\dot{\eta}(t) &= A_1\eta(t) + B_1v(t) \\ y(t) &= C\eta(t).\end{aligned}\tag{10}$$

- Basically, it is an approximation to the original nonlinear model in the region X_1 .



Feedforward (LQ) Controller (Cont'd)

- To zero out steady-state error when tracking constant signal r , we introduce an integral action and define $e(t) = \int_0^t (r - C\eta(t))dt$ and

$$\bar{\eta}(t) = \begin{bmatrix} \eta(t) & e(t) \end{bmatrix}^T.$$

- We have an augmented system:

$$\begin{aligned} \dot{\bar{\eta}}(t) &= \bar{A}_1 \bar{\eta}(t) + \bar{B}_1 v(t) + B_r r, \\ y(t) &= \bar{C} \bar{\eta}(t) \end{aligned} \tag{11}$$

- Next, discretize system (11) and apply the discrete-time Linear Quadratic (LQ) optimal control technique to determine the feedforward controller.

Linear Quadratic Regulator (LQR)- Revision

- Problem: Consider the discrete-time system:

$$\eta(k+1) = \Phi\eta(k) + \Gamma v(k)$$

and the cost function $J = \frac{1}{2} \sum_{k=0}^{\infty} (\eta^T(k) Q \eta(k) + v^T(k) R v(k)).$ (12)

The objective is to find the feedback control sequence $\{v(k)\}$ that minimizes J for given $Q \geq 0$ and $R > 0$ (Infinite-time horizon LQ).

- Solution: The minimizing control law is

$$v(k) = -L\eta(k), \quad L = (\Gamma^T S \Gamma + R)^{-1} \Gamma^T S \Phi \quad (13)$$

where S is the solution to the Algebraic Riccati Equation (ARE)

$$S = \Phi^T [S - S \Gamma (\Gamma^T S \Gamma + R)^{-1} \Gamma^T S] \Phi + Q. \quad (14)$$

Linear Quadratic Regulator (LQR)-Revision (Cont'd)

- The cost function:

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (\eta^T(k) Q \eta(k) + v^T(k) R v(k)). \quad (15)$$

- Different $Q \geq 0$ and $R > 0$ in (15) will result in different stable closed-loop poles for $(\Phi - \Gamma L)$.
- Choose Q and R to yield a desired performance.
- By observing (15), the first term relates to *energy in the system*; whereas the second term relates to *energy consumed by the controller*. Therefore:
 - Adjust R to meet constraints on the control input, e.g. expenditure of control;
 - Adjust Q to meet desired performance criteria, e.g. settling time.

Feedforward (LQ) Controller

- To apply the discrete-time LQR technique for the FF controller, we discretize the augmented system (11) with a sampling period T , we then have

$$\begin{aligned}\bar{\eta}(k+1) &= \Phi\bar{\eta}(k) + \Gamma v(k) + \Gamma_r r, \\ y(k) &= \bar{C}\bar{\eta}(k).\end{aligned}$$

- Then apply the discrete-time LQR technique to (29) with the cost function

$$J = \sum_0^{\infty} (\bar{\eta}'(k)\bar{Q}\bar{\eta}(k) + v'(k)Rv(k)), \quad Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & q \end{bmatrix}, \quad q > 0, \quad R > 0$$

and set $r = 0$.

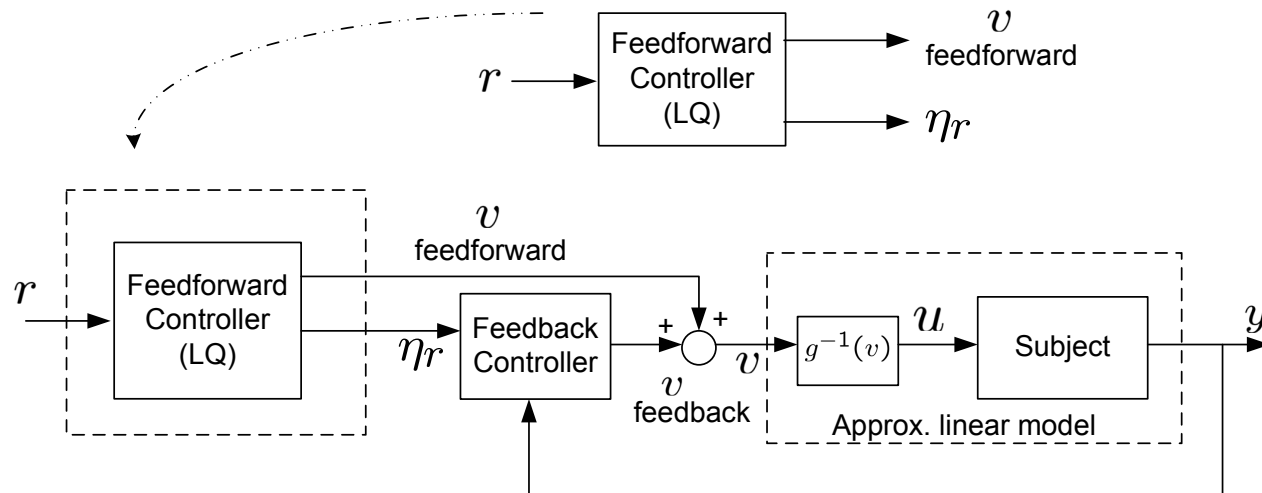
- We obtain a gain L that gives the feedforward control controller $v(k)_{ff} = L\bar{\eta}(k)$.

Feedforward (LQ) Controller (Cont'd)

- The LQ feedforward controller is in the form:

$$\begin{aligned}\bar{\eta}(k+1) &= \Phi\bar{\eta}(k) + \Gamma v_{ff}(k) + \Gamma_r r, \\ v_{ff}(k) &= L\bar{\eta}(k) \\ \eta_r(k) &= \bar{C}\bar{\eta}(k),\end{aligned}\tag{16}$$

r is a given reference input. The signals η_r and $v_{ff}(t)$ will be *pre-computed*.



Feedback Controller Design: H_∞ Control

- To cope with the uncertainty in the model, we design an H_∞ feedback controller. Since the model is nonlinear $\dot{\eta} = f(\eta, v)$, we linearize it at $\eta_0 = [0.5 \ 0.115]^T$, $v_0 = 0.452$, and

$$\dot{\xi}(t) = \frac{\partial f}{\partial \eta} \xi(t) + \frac{\partial f}{\partial v} w(t)$$

and hence the linearized model is

$$\begin{aligned}\dot{\xi}(t) &= A_2 \xi(t) + B_2 w(t) \\ \zeta(t) &= C \xi(t)\end{aligned}\tag{17}$$

where $\xi(t) = \eta(t) - \eta_0$ and $w(t) = v(t) - v_0$.

- The transfer function of the linearized model is given by

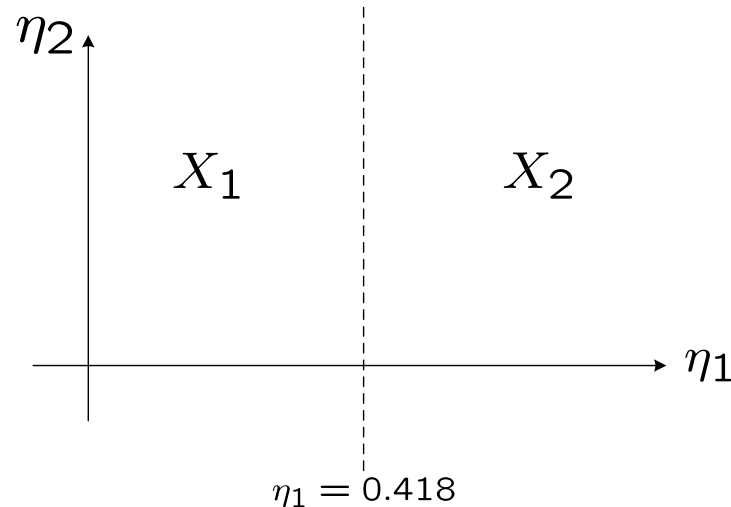
$$G(s) = \frac{0.027s + 2.86 \times 10^{-5}}{s^2 + 0.0317s - 8.65 \times 10^{-6}}.\tag{18}$$

Feedback Controller Design: H_∞ Control (Cont'd)

• The linearized model

$$\dot{\xi}(t) = A_2\xi(t) + B_2w(t); \quad \zeta(t) = C\xi(t)$$

is an approximation to the original nonlinear model in the region X_2 .



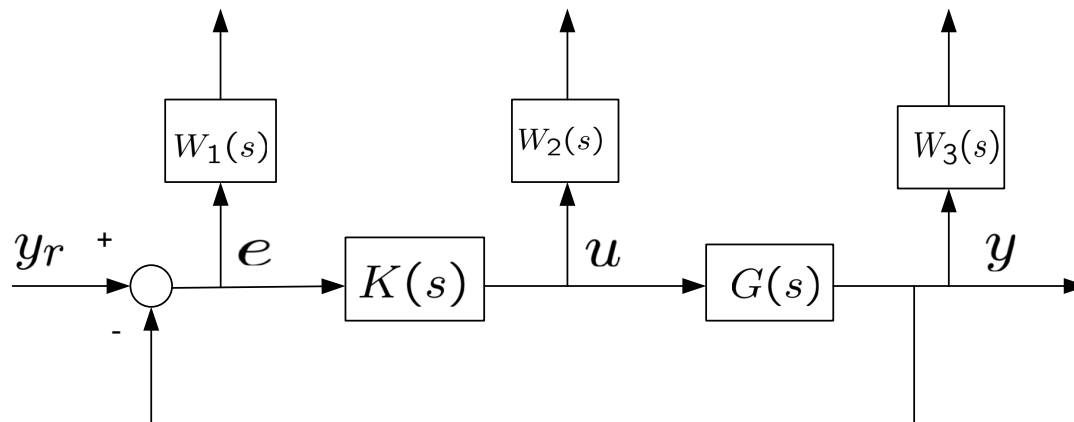
- To design the H_∞ control, we formulate the control problem as a mixed sensitivity problem.

Feedback Controller Design: H_∞ Control (Cont'd)

- In a mixed sensitivity problem, the idea is to choose some weighing functions ($W_1(s)$, $W_2(s)$ and $W_3(s)$) defined by the control objectives and to find a controller $K(s)$ such that:

$$\left\| \begin{bmatrix} W_1(s)S(s) \\ W_2(s)R(s) \\ W_3(s)T(s) \end{bmatrix} \right\|_\infty = \left\| \begin{bmatrix} W_1(s)(I + G(s)K(s))^{-1} \\ W_2(s)K(s)(I + G(s)K(s))^{-1} \\ W_3(s)G(s)K(s)(I + G(s)K(s))^{-1} \end{bmatrix} \right\|_\infty \leq \gamma$$

where $S(s)$ and $T(s)$ are the sensitivity and complementary sensitivity, respectively. (Loopshaping design)



Feedback Controller Design: H_∞ Control (Cont'd)

- $W_1(s)$ is chosen as a high-gain lowpass filter approximating an integral action in order to ensure good tracking accuracy.
- A first-order high-pass filter is chosen for $W_2(s)$ to limit the controller bandwidth and magnitude.
- The weighting function $W_3(s)$ is chosen to accommodate any multiplicative modeling error.
- The weighting functions were chosen as

$$\begin{aligned}W_1(s) &= \frac{0.1(s + 0.083)}{(s + 8.33 \times 10^{-5})} \\W_2(s) &= \frac{70(s + 6.25 \times 10^{-4})}{(s + 4.38)} \\W_3(s) &= \frac{100(s + 0.066)}{(s + 8.33)}.\end{aligned}\tag{19}$$

Feedback Controller Design: H_∞ Control (Cont'd)

- We then obtain a robust controller

$$K(s) = \frac{8.91 \times 10^{-3}s^4 + 0.11s^3 + 0.33s^2 + 0.01s + 2.6 \times 10^{-6}}{s^5 + 9.1s^4 + 6.43s^3 + 0.42s^2 + 4.61 \times 10^{-4}s + 3.56 \times 10^{-8}}.$$

- The controller gives $\gamma = 0.9$ so that, for all $\omega \in \mathbb{R}$,

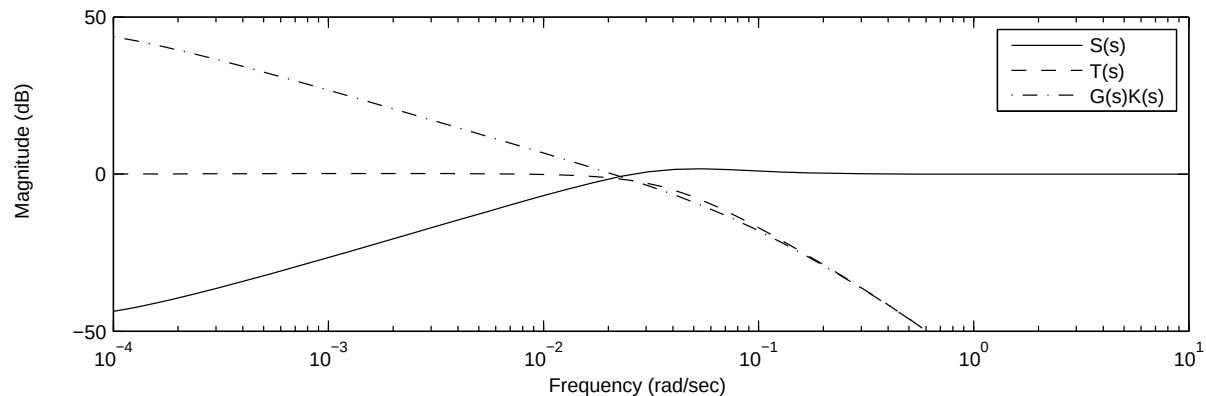
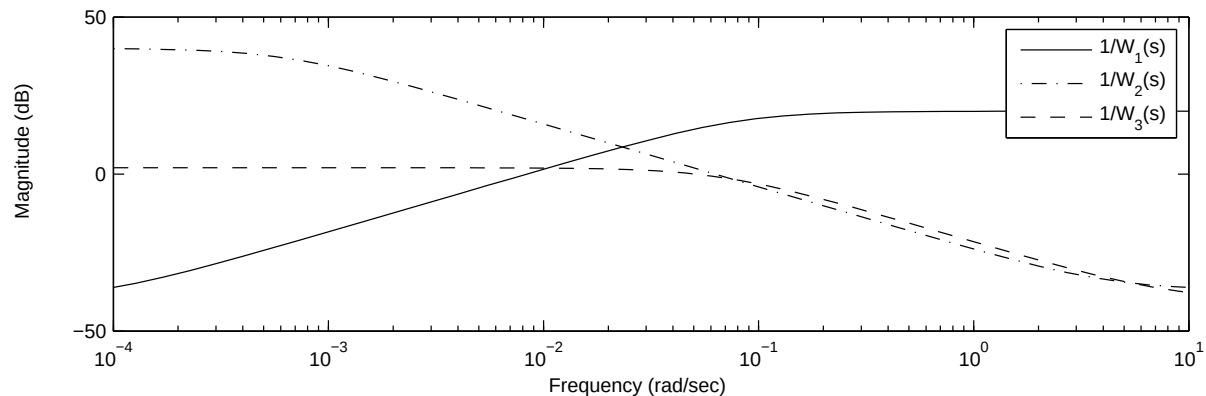
$$|S(j\omega)| \leq \gamma/|W_1(j\omega)| < 1/|W_1(j\omega)|$$

$$|T(j\omega)| \leq \gamma/|W_3(j\omega)| < 1/|W_3(j\omega)|$$

$$|K(j\omega)S(j\omega)| \leq \gamma/|W_2(j\omega)| < 1/|W_2(j\omega)|.$$

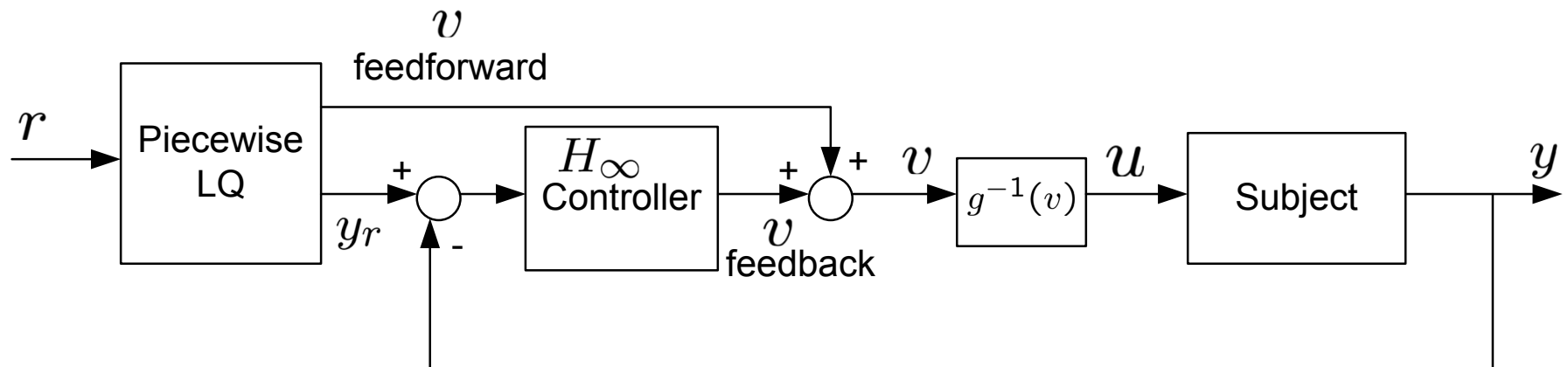
Feedback Controller Design: H_∞ Control (Cont'd)

- (TOP) The inverse of the weighting functions $W_i(s)$, $i = 1, 2, 3$.
- (BOTTOM) Sensitivity $S(jw)$, complementary sensitivity $T(jw)$, and loop gain $G(jw)K(jw)$.



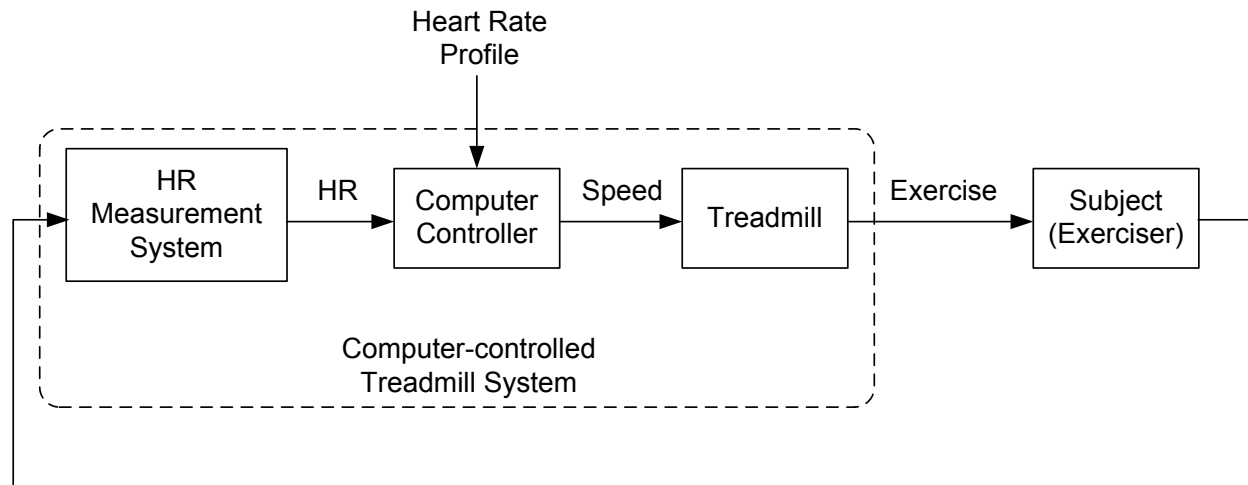
Overall Controller

- Overall controller: feedforward + feedback controllers:



Computer-controlled Treadmill System

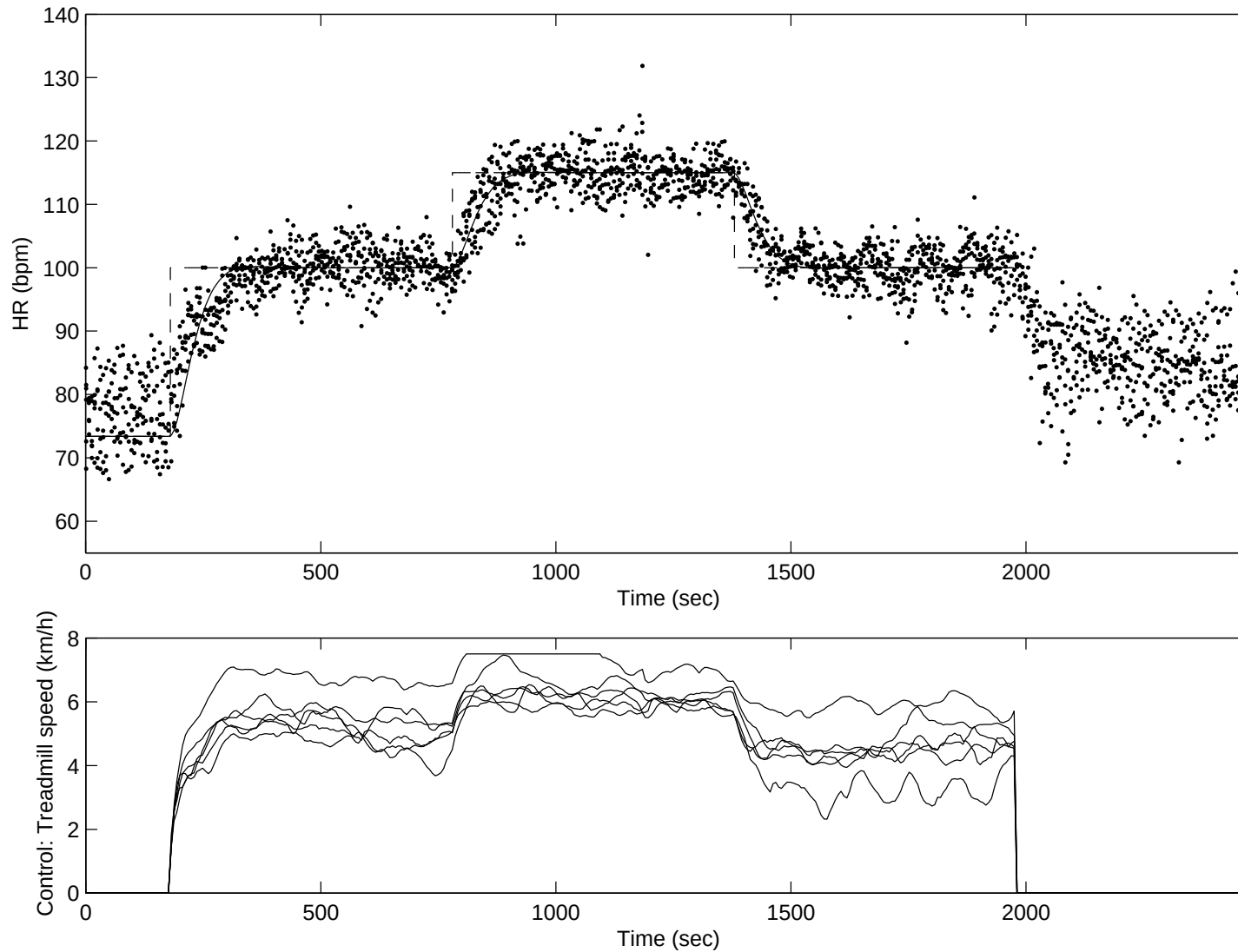
- Using the control design, a computer-controlled treadmill system was implemented for HR regulation. The system consists of 3 main components: a treadmill, a heart rate measurement system, and a computer controller.



- The computer controller was implemented in LabVIEW.
- Feedforward controller was pre-computed offline.

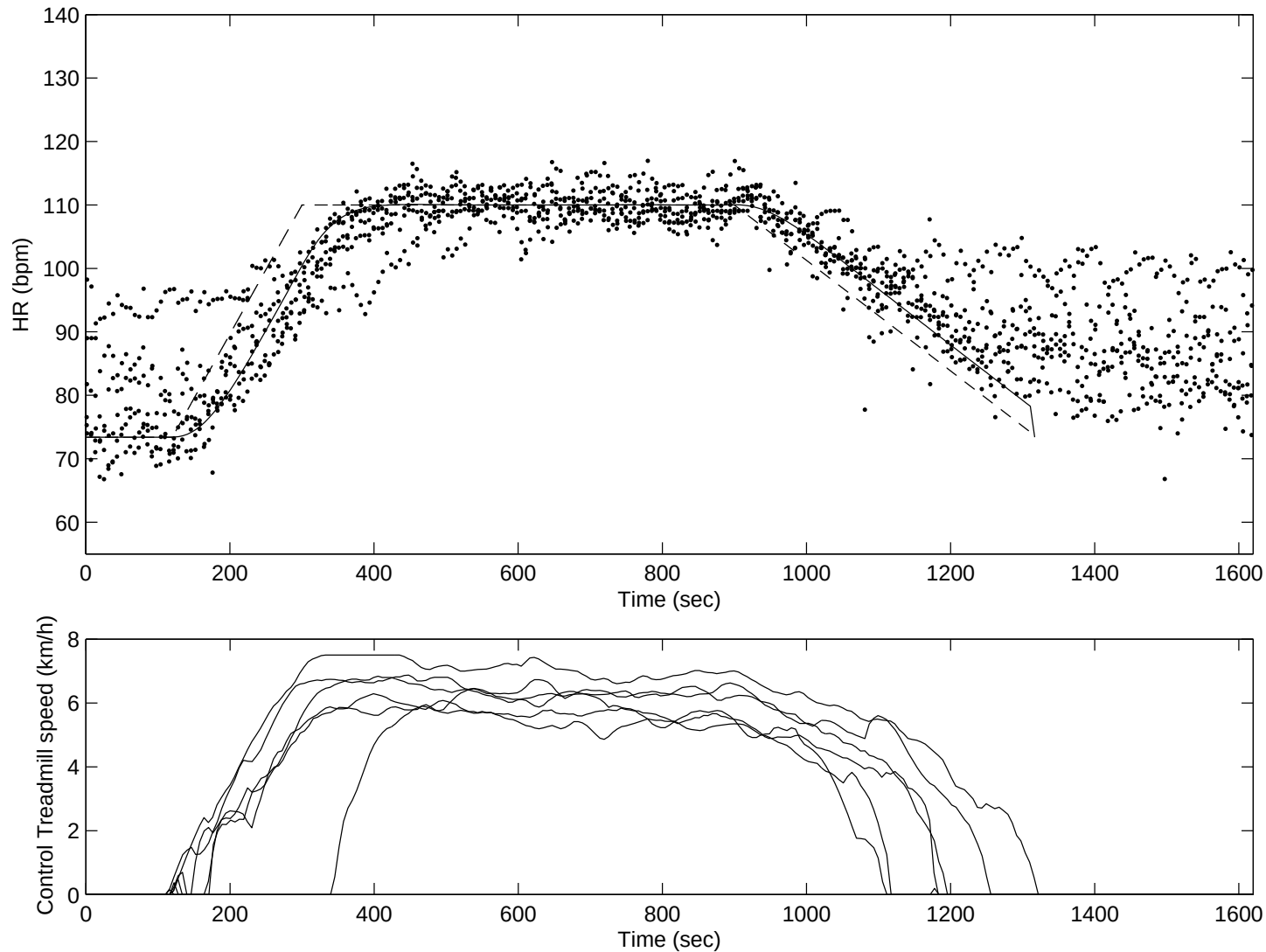
Control Results

Regulation of heart rate at 100 and 115 bpm for all 6 subjects.



Control Results (Cont'd)

Heart rate regulation at 110bpm for all 6 subjects with gradually warm-up and cool-down periods.



Concluding remarks

- A **nonlinear dynamic model** describing the HR response to the treadmill walking exercise was proposed. The feedback interconnected structure of the model would be useful in describing the macro behaviors of the central and the peripheral local responses to exercise.
- A **computer-controlled treadmill system** was implemented for the regulation of HR during treadmill walking exercise. The controller consists of a LQ feedforward and an H_∞ feedback controllers.